

Title

Declining Arable Land Per-Capita and Increase Food Insecurity in Pakistan: A Data Driven Structural Analysis

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1. Introduction

Food security is a critical social and economic issue worldwide, particularly in developing and underdeveloped economies. Climate change has made it ever more severe due to unpredictable rainfall, floods, and multifaceted environmental issues. Literally, there are three main causes of food insecurity: a rapid increase in population, degradation in agriculture land and a decrease in per capita arable land. Food security in Pakistan is a never-ending problem that has historically become worse. Statistically, in 1991, around 352,700 sq. km of land was arable, with the total population around 120 million, and by the end of 2025, the population surged to 255.2 million and arable land decreased to 352,360 sq. km. Broadly, the population increased by 113 per cent, and alternatively, agricultural land decreased by around 0.1 per cent. This alarming situation could become clearer, as Thomas Malthus proposed the Malthusian Theory of Population and stated that when population grows exponentially, comparatively, the agricultural capacity of the country increases non-linearly or even remains stagnant; this adverse relationship causes resource crises. This scarcity of agriculture resource causes food insecurity.

The food security issue is about the production of crops and their usage at the local level to produce enough food to meet the domestic need. Keep in view the present time population and growth rate as

well. Because, in recent times, arable land has remained fixed for the last few years; alternatively, population is constantly increase with 1.50 % per annum. In a nutshell, this creates an issue of pressure gap that measures food security numerically. Pressure gap is measured by the difference between arable land per capita and population percentage change. This study investigates the food security issue of Pakistan using the pressure gap self-calculated measure. The primary objective of this study is to understand the cyclic trends and movements over time. And, secondary purpose is to build a narrative based on data-driven inferences to generalise this information for greater social change. This research helps policymakers and thinkers understand the real story in simple terms, distilled from the numbers.

2. Methodology and Analysis

2.1. Dataset

This study is based in Pakistan and incorporates secondary quantitative data collected from the World Bank website from 1991 to 2025. These factors include Total Population, Population Growth (Annual %) and Agricultural Land (sq. km). In addition, Agricultural Land (sq. km) percentage change was calculated using the change formula. Agricultural Land Per Capita (ha/person) is also an author-calculated measure. At last, the main factor, Pressure Gap, is measured via manual calculation in Excel. The following are the main numerical calculations.

(a) Agricultural Land (sq. km) %age Change

$$\text{Agricultural Land (sq. km) \%age Change} = \frac{\text{Current Agricultural Land (sq. km)} - \text{Previous Agricultural Land (sq. km)}}{\text{Previous Agricultural Land (sq. km)}} \times 100$$

(b) Agricultural Land Per Capita (ha/person)

$$\text{Agricultural Land Per Capita (ha/person)} = \frac{\text{Total Agricultural Land in Hectaes}}{\text{Total Population}}$$

(c) Pressure Gap (Food Insecurity)

$$\text{Pressure Gap (\%)} = \text{Percentage Change in Population} - \text{Percentage Change in Agriculture Land}$$

2.2. Variables Pre-processing

New Index Name	Purpose	Formula
Population Index	Shows how population changed compared to 1991	$(\text{Total Population in Year } t / \text{Total Population in 1991}) \times 100$
Agricultural Land Index	Shows how agricultural land changed compared to 1991	$(\text{Agricultural Land in Year } t / \text{Agricultural Land in 1991}) \times 100$
Agricultural Land Per Capita Index	Shows how land availability per person changed compared to 1991	$(\text{Agricultural Land Per Capita in Year } t / \text{Agricultural Land Per Capita in 1991}) \times 100$

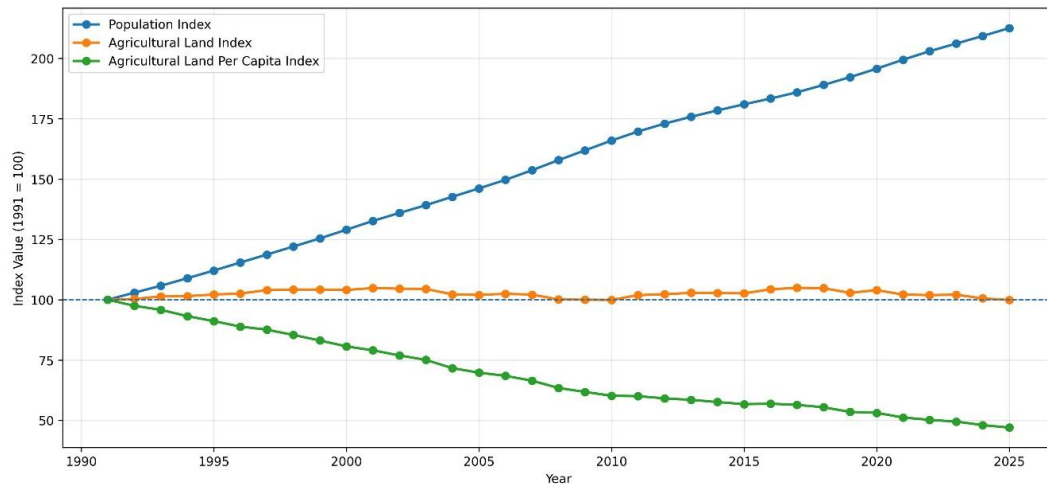
Pre-processing is incorporated to standardize that data and make the scales equal. It helps to produce comparable results for ensuring the generalizability. These indices are created to transform all factors into a common base across all years, with the value 100 for the base year 1991. Theoretically, population-resource pressure provides a fundamental basis. So, these indices provide a comprehensive and clear comparative study of the arable land per capita, population, population growth and total agricultural land. This modest approach makes it structurally useful for measuring food insecurity over time in Pakistan.

2.3. Analysis and Visualisations

Figure 1. Structural Food Security Pressure in Pakistan: Indexed Trends of Population, Agricultural Land, and Land Availability (1991–2025)

Q1. How does index-based analysis help to understand structural issue compressively and numerically?

Q2. What are the trends of population growth, arable land and arable land per capita?

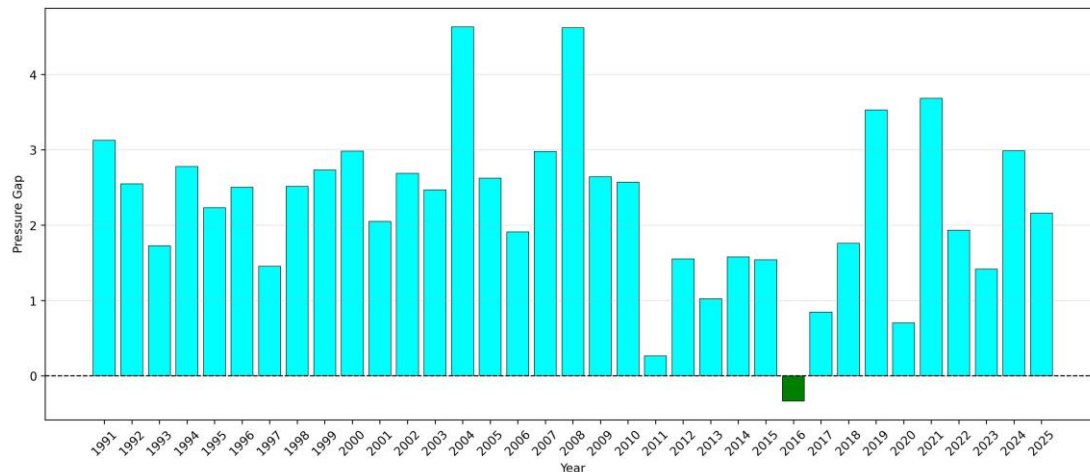


The index-based analysis of structural food insecurity in Pakistan covers the period of 1991 to 2025 and presents the direction comparison of all three indices. The population increased consistently, and it became double as compare to the 1991 numbers. Alternatively, the arable land index remained flat for all years, showing that land did not expand to produce food-related raw materials. In addition, agricultural land per capita gives a comprehensive view, indicating that land available per person decreases over the years. Simply, Pakistan is feeding the doubling population with the same arable land. These findings are supported by the Malthusian population-pressure hypothesis: increasing population growth exerts pressure on food and agriculture resources. Additionally, the carrying-capacity hypothesis states that arable land is a finite resource that is hard to expand, but on the other hand, the population keeps increasing. This analysis suggests that the food insecurity issue of Pakistan is a long-term imbalance issue of resource management, population control and weak agriculture governance and planning.

Figure 2. Population–Land Pressure Gap and Food Security Stress in Pakistan (1991–2025)

Q1. What is the association between population growth and arable land?

Q2. What are the trends of the food insecurity (Pressure Gap) across 35 years?



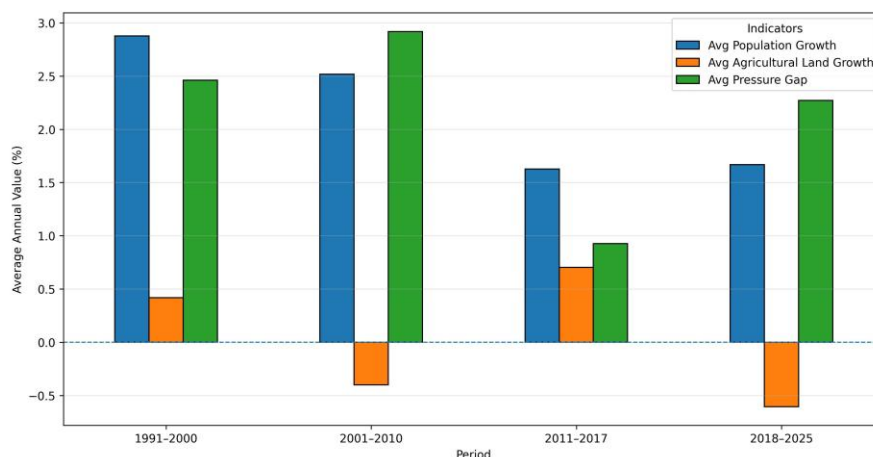
The

pressure-gap remains positive in all years except 2016, indicating food security is an enduring critical issue since 1991. Arable land and agricultural growth remain stagnant, but population growth keeps increasing with an average rate of 1.5% in all years. It's a simple phenomenon: when population grow food demand increases, but production of raw materials did not grow or even maintain the balance. This analysis suggests that food insecurity in Pakistan is not a short-term problem, but would become even more hazardous in the future due to climate change and environmental pressure on the agriculture sector.

Figure 3. Period-Wise Food Security Pressure Regimes in Pakistan (1991–2025)

Q1. In which historical time period was structural food insecurity highest?

Q2. What are the trends of the average arable land, population growth and arable land per capita across the years?



The analysis helps to understand the overall structural and systemic issues of food insecurity. Period-wise food security did not remain constant, showing positive and critical negative trends. During the 1991-2000 period, population increased significantly, and arable land growth remained modest, which created a bearable pressure gap. The 2001-2010 remain the most critical because arable land growth showed negative trends, which increased the pressure gap significantly. During the years 2011-2017, the pressure gap started improving due to positive trends in arable land growth and population increase. Lastly, the years 2018-2025 again found negative trends in arable land growth, which increased the pressure gap. This analysis critically observed that the food insecurity issue is twofold: population increase and stagnant or decreasing arable land. Additionally, there are other socio-economic, environmental, technological and financial issues that also moderate or mediate this relationship. This is a structural phenomenon of numerous interlinked factors that need to be investigated to produce a declarative policy suggestion to address the Pakistan food insecurity issue.

3. Limitations and Delimitations

3.1. Limitations

1. This study is limited to two factors, which might give a narrow perspective on the food security issue.
2. This study does not include the other factors related to socio-economic, environmental, technological and financial aspects that might moderate or mediate this relationship.

3.2. Limitations

1. The delimitation of this study is only two factors; the main purpose is to understand the two main natural resource associations for food insecurity.
2. It is based on simple and easy analysis just to provide a comprehensive understanding of the food insecurity issue.

4. Conclusion

This research found that arable land and population growth are two contributing factors in Pakistan. It also observed that the food insecurity issue has become a real national issue and it requires a comprehensive and urgent remedy. A very critical and dangerous observation of this study is that the food security issue has lingered since 1991, and it has become more hazardous in recent times. This study also observed that arable land growth went negative in the period 2001-2010 and 2018-2025, and in the same cycles, the pressure gap on food security became even worse. This study aimed to provide a comprehensive understanding and to build a narrative on social issues based on data-driven decision-making. In future, this study suggests including socio-economic, environmental, technological and financial aspects to build a better understanding and provide clear policy suggestions.

5. Policy Suggestions

1. Governments and institutions must adopt the technology in agriculture to increase the production keeping via the fundamentals of the carrying-capacity hypothesis, because it suggests arable land is a finite resource.
2. There must be an effective population control and management policy, and it must be implemented, satisfying all the Islamic religious values. Additionally, religious actors and pressure groups must be controlled and restricted to mosques and madrasas, and play their true role in the proliferation of Islam and the protection of true Islamic values based on Sunna and Quran. Because its social and biological issue, it must be addressed by experts from that domain.
3. The government must provide a dedicated focus on climate adaptation and resilience efforts at the local level. Because climate change is a global issue and its keep increasing

in the future as well, so only addressable issue is adaptation and rehabilitation at the regional, domestic and local level. This increases the productivity and variety of agricultural and horticultural crops.

4. Government must truly and urgently functionalize the local government system, because agriculture is a rural subject; Pakistan's rural population is around 62% in Pakistan, so the local government system helps to address agriculture issues at farmers' doorstep and on time.
5. The government must abolish all unrelated and white-elephant bodies; comparatively, it must functionalize only true and field-expert regulatory bodies at the national, provincial, regional and district level to effectively channel their issues and their addressed on time.